

Grounding the Decision Constants of a Self-Evaluating Research System

A sourcing proposal: which values the literature determines,
which it merely contains, and which it does not touch

Prepared with `opencode`

Companion to *Provisional Constants* (`paper.pdf`) and Figure 1 (`flow.pdf`)

Audited system: `github.com/cloudcell/labloop`, commit 21d91f1

Abstract

An audit of an agentic ML research system found fifteen numeric constants that gate scientific decisions and carry no stated derivation. This document attempts, for each, to locate literature that *determines* the value. Four are groundable in a specific source; two are settled by a theorem rather than by a number; nine have no literature basis and should either be reclassified as operational preferences or deleted. The largest single result is negative: a constant producing a bounded score for a ratio whose denominator may be near zero is *invalid*, not merely conservative, by a theorem of 1987. The second is quantitative: a session size of $n \approx 9$ seeds per arm is required for 80% directional accuracy at a specified smallest effect size of interest, a figure the audited system's own benchmark independently confirms to within 0.023, against a shipped default of $n = 1$.

1 Purpose

The companion paper argues that a decision constant is not a parameter but a claim, and that an unsourced claim should be labelled `PROVISIONAL` rather than quietly published. This document performs the labelling. For every decision constant identified in the audit, it asks a narrow question: *is there a retrievable source that fixes this value, and states why?* Where the answer is no, that is recorded as a finding, not as a failure to search hard enough.

The result is stated up front so that the tables are not read as more favourable than they are.

Of fifteen decision constants audited, four can be grounded in a specific citable source, two are settled by a mathematical theorem rather than by a value, and nine have no literature basis at all — and should be reclassified as operational, or deleted.

A large fraction of the numbers that attracted audit attention are not epistemic claims. They are timeouts, retention counts and bind addresses. Presenting those to a literature review is a category error, and part of this proposal's purpose is to say so explicitly rather than to manufacture citations for them.

2 What counts as grounding

A constant is `SOURCED` only if a retrieved source states the value *and* the reason for it, such that a maintainer could substitute a different value knowing what breaks. Three weaker outcomes are recorded separately and must never be reported as grounding.

Table 1: Status vocabulary. The third row is the failure mode this proposal is most at risk of.

Status	Meaning
<code>SOURCED</code>	A source names this value, or a procedure that yields it, with justification.
<code>DERIVED</code>	No source gives the value, but a method does; the value follows from inputs we supply.
<code>IN-RANGE</code>	A source tabulates a family of values and ours is one of them, but the source does not single ours out.
<code>UNGROUNDED</code>	No source found. Delete, reclassify as operational, or retain as <code>PROVISIONAL</code> with a stated reason.

`IN-RANGE` is separated because it is how an audit of this kind deceives itself: a paper is found that happens to contain the number 0.3 somewhere in a table of unrelated quantities, and the constant is declared grounded. The distinction is maintained throughout §6.

3 C1 — Calibration constants

3.1 The prior ceiling: 0.3

This is the best-anchored constant in the system. The National Academies’ report on reproducibility [1] tabulates the posterior probability that a hypothesis is true as a function of the prior probability and the observed p -value. Its Table D-1 gives, for a one-tailed $p = 0.05$:

Table 2: Posterior probability that the hypothesis is true, one-tailed $p = 0.05$ (National Academies 2019, Table D-1). The bold column is the audited system’s prior ceiling.

prior $P[H_1]$	.01	.05	.10	.20	.25	.30	.40	.50
posterior	.038	.169	.301	.492	.563	.624	.721	.795

The report discusses 0.3 explicitly as “*its pre-experimental probability of being true is about 1 in 3.*” That is a substantively defensible prior for a scientific claim, and it maps cleanly onto a ceiling for a claim with no evidence-bearing edge: an ungrounded claim is worth your prior, and one-in-three is a documented choice rather than a guess.

Proposal. Retain 0.3, but cite the source at the definition site, make the assumed prior a deployable configuration value, and look the ceiling up in the same table rather than hardcoding it. An operator who believes their prior is 0.10 should obtain a ceiling near 0.30, not 0.3.

Caveat. The report offers 0.3 as one row among many, not as a recommendation. The status is `IN-RANGE`, not `SOURCED`.

3.2 Verdict confidence: 0 . 85

This is the constant the literature actively indicts rather than merely failing to support. From the same table, at a one-tailed $p = 0.05$ the posterior *cannot exceed* 0.795 even at a 50% prior. The audited system mints 0.85 on a verdict that may rest on a single trial, with no p -value computed and no evidence count recorded. Reaching 0.85 requires simultaneously a high prior and a strong result; the system assumes the first by fiat and never measures the second.

The report also states the principle the system violates [1]:

“It is clearly inappropriate to apply the same confidence to the results of a study with a highly unexpected and surprising result as in a study in which the results were a priori more plausible.”

The system applies an identical confidence to every accepted and every rejected hypothesis, regardless of effect size, variance, or surprise — which is the specific thing the sentence forbids.

Proposal. Replace the verdict-to-float lookup with a computed posterior using the deployment’s declared prior and an effect-size test. For the common case of prior 0.3 at one-tailed $p = 0.05$, Table D-1 gives 0.624. Where no p -value is available, the correct output is an enumeration, not a float. A further check the system currently lacks is available: the Brier score decomposes into *calibration* and *refinement* components [11], so the system could audit whether its confidences are calibrated rather than assert that they are.

3.3 Meta-decision confidence: 0 . 6

No literature supports 0.6, and the provenance trace is worse than merely unsourced. The value is traceable to a unit-test fixture whose assertion is that the value *must be rejected*:

```
r = await call_tool(mcp, "record_finding", {
    "investigation_id": inv_id, "content": "too confident",
    "confidence": 0.6,
})
assert "prior ceiling" in r["error"]
```

(tests/zetesis/test_20260916-1721Z-enforcement.py:109)

The production call site sits in `arete/tools/decisions.py:122` beside a comment explaining that evidence edges are minted inline because the claims server caps unevidenced claims at the prior ceiling. The mechanism is documented; the magnitude is not.

Proposal. Delete. Replace with the same posterior computation as above, keyed on the decision’s own evidence count. Meta-decisions are required to cite at least one evidence reference, so the inputs exist — they are simply never read.

3.4 Importance and edge weight

`edge.weight = 1` is defensible *if* it is read as *unweighted*, which is the honest interpretation: the claim graph does not attempt to rank evidence. The difficulty is that a field named *weight* carrying a constant invites the reading that evidence is being weighted when it is not. Rename it to declare the fact, or drop it.

`importance = 0.5` has no defensible reading. It is the midpoint of the unit interval and it responds to nothing.

Proposal. Delete importance. No citation is needed to delete a field that carries no information — and that is a better outcome than a spurious reference.

4 C2 — Discriminative constants

4.1 Zero-divisor handling: settled by theorem

This is the one item where the literature supplies no number but *proves the current implementation wrong*. The proof has been available since 1954.

Fieller’s construction of a confidence interval for a ratio of two means yields an *unbounded* interval whenever the denominator is not significantly different from zero [2] — either the whole real line, or two disconnected infinite rays. This is not a defect of the method; it is the correct answer, since a ratio whose denominator is compatible with zero can be arbitrarily large in either direction.

Gleser and Hwang [3] then close the door on the alternative. A retrieved restatement gives their result as a theorem:

“Any confidence interval which has finite length for almost every observation has minimum zero coverage probability while minimizing over (α_1, α_2) .”

Read plainly: *one cannot manufacture a bounded, valid score for a ratio when the denominator may be near zero*. Any method that always returns a finite number in that situation has zero coverage. It is not a conservative estimate; it is an invalid one. Returning 0.0 is a decision that has been taken, not a limit that has been reached. Independent reviews reach the same conclusion and note in addition that the naive ratio estimator is biased away from the singular case, so the problem is not confined to exact zero [4, 5].

Empirically this is what was observed. One campaign closed with a frozen score of 0 for `champion_mean = 0` against `challenger_mean = 0.5`: the challenger beat the champion, the ratio is undefined, and the frozen score is the worst value the scale admits. The same session produced a tournament, driven with the identical input, that correctly refused to close.

Proposal.

1. The campaign closer must *refuse* a zero divisor, matching the tournament closer, which already does. This is a one-line correctness fix and the single highest-value change identified by the audit.
2. Replace the point ratio with a Fieller confidence set and record the set’s boundedness. The correct output for a near-zero denominator is *undefined*.
3. Refusing on exact equality to zero catches only the measure-zero case. Gate on the Fieller condition instead: the denominator’s confidence interval must exclude zero. That condition covers the whole neighbourhood, which is where the 37.8% sign-error rate lives.

4.2 Session size n : derived from a citable procedure

This is the constant with a fully specified literature procedure, and the procedure is unambiguous: declare a smallest effect size of interest, then compute the n that achieves a stated power against it [8, 9]. Two warnings from the same literature apply directly.

First, do not use the observed effect size from a pilot to set n . Albers and Lakens [9] demonstrate that pilot-derived effect sizes are upwardly biased by follow-up bias, which systematically yields underpowered main studies. The audited system’s tournaments routinely run one seed and record what they observe; that is the pilot-as-design error.

Second, relying on a benchmark is “the weakest possible justification” of a smallest effect size of interest [8]. The threshold must come from the individual contract, not from a global default.

4.2.1 Worked example, validated against the system’s own data

Using the parameters of the audit benchmark: base accuracy $\mu = 0.50$, per-observation $\sigma = 0.05$, and a smallest effect size of interest $\delta = 0.02$ in the challenger-minus-parent difference. On the log-ratio scale $\delta \approx \log(0.52/0.50) = 0.03922$, and the delta-method standard error is $\sigma\sqrt{2/n}/\mu$. Solving for the n giving 80% directional accuracy ($z = 0.8416$):

$$n = 2 \left(\frac{\sigma z}{\mu \delta} \right)^2 = 2 \left(\frac{0.05 \times 0.8416}{0.50 \times 0.03922} \right)^2 = \mathbf{9.2} \text{ seeds per arm} \quad (1)$$

and $n \approx 21.4$ per arm for 90%.

The normal approximation is not a hand-wave: the audited system’s own Monte-Carlo benchmark confirms it to within 0.023 at every sample size tested (Table 3).

Table 3: Closed-form prediction of sign error against measured Monte-Carlo sign error, 400 replicates per cell, $\sigma = 0.05$, true effect +0.02.

seeds / arm	predicted	measured	error
1	0.391	0.378	+0.013
5	0.268	0.245	+0.023
20	0.107	0.103	+0.004

So the figure of roughly nine is not a formula transplanted from a textbook; it is a prediction the audited system’s own recorded data supports. The system ships with a default of $n = 1$ — a nine-fold deficit — and records neither the required nor the achieved n .

Proposal.

1. The promotion policy, currently inert, must require a smallest effect size of interest and a target power, and the tournament must refuse to open without them. This is what makes the field non-inert.
2. Store $n_{\text{requested}}$, n_{achieved} , the effect size of interest, and the Type S / Type M error [7] on the frozen score row itself. A score is uninterpretable without them.

4.3 Promotion threshold: an odds ratio, not a metric ratio

The current decision is $\text{score} > 1$ — an unweighted ratio with no threshold search. The Bayes-factor literature supplies a graded evidence scale better suited to the task, with named rungs a contract may select from [10]:

Proposal. Replace the bare threshold with a minimum-evidence requirement expressed on this ladder, and require the contract to name which rung constitutes promotion. The value is contract-specific and cannot be defaulted.

Table 4: Evidence rungs on the $2 \ln BF$ scale (Kass & Raftery 1995).

$2 \ln BF$	0–2	2–6	6–10	10–150	> 150
reading	not worth more than a bare mention	positive	strong	very strong	decisive

5 C3 and C4 — Governance and operational constants

5.1 Staleness, freshness, epoch: reclassify, do not cite

No literature was found for 3600 (staleness), 600 (freshness time-to-live), or 86400 (improvement epoch), and I do not believe any exists. These are not scientific claims. A staleness threshold states how long an operator will wait before treating an open record as abandoned; it is a property of the deployment’s cadence, not of nature.

The correct action is therefore **reclassification, not citation**: these belong in the governance and operational classes, and the disclosure should say so rather than list them beside a verdict confidence.

Two of them do carry a defect, but it is a duplication defect rather than a missing one. The 600 freshness value is written seven times across four packages, so setting it in one place does not set it in the others. Log retention is 100 files in four packages and 30 in one, meaning the audit trail that gates all writes has different retention in one server than in the others. Both are fixed by a single shared constants module and a stated reason, and neither requires a citation.

5.2 Operational remainders

The observation grace period, the stalled-trial margin, and the 1.0-second margin that absorbs timestamp rounding are likewise ungrounded, and should be recorded with a reason string rather than a reference: the last is a fact about the clock, not about science. The default bind address is a security-posture decision and belongs in an operator knob.

6 Summary

7 Acceptance criteria

The proposal is implemented when:

1. **R1** — every constant carries a source status from §6 in both the code and the API schema.
2. **R2** — no constant is defined in more than one location; a continuous-integration check fails on the second definition.
3. **R3** — the campaign closer refuses a zero divisor, matching the tournament closer.
4. **R4** — a tournament cannot open without an effect size of interest and a target power in its promotion policy; the field stops being inert.

Table 5: All fifteen decision constants, with sourcing status and proposed action.

Constant	Value	Status	Proposed action
prior confidence ceiling	0.3	IN-RANGE	keep; cite source; make prior deployable
verdict confidence	0.85	UNGROUNDED	replace with posterior; 0.624 for prior .3, $p=0.05$
meta-decision confidence	0.6	UNGROUNDED	delete; compute from evidence count
claim importance	0.5	UNGROUNDED	delete — carries no information
edge weight	1	UNGROUNDED	rename to declare unranked, or drop
zero-divisor fallback	0.0	VOID	refuse; Fieller / Gleser–Hwang
seed count	1	DERIVED	$n \approx 9$ per arm (80%); SESOI in contract
promotion threshold	> 1	DERIVED	minimum rung on the Bayes-factor ladder
staleness	3600	ungrounded	reclassify; deduplicate
freshness TTL	600	ungrounded	reclassify; deduplicate (7 copies)
improvement epoch	86400	ungrounded	reclassify
observation grace	86400	ungrounded	reclassify
stalled margin	300	ungrounded	reclassify
log retention	100 / 30	ungrounded	unify; document as audit policy
bind address	0.0.0.0	posture	operator knob

5. **R5** — the frozen score row carries requested and achieved sample size, the effect size of interest, and the sign and magnitude errors where computable.
6. **R6** — confidence is either computed from a stated prior and likelihood, or is an enumeration.
7. **R7** — constants with status `UNGROUNDED` may not be cited as grounds for a promotion decision. This is the rule that makes the table above binding rather than decorative.
8. **R8** — a generated disclosure table — value, file, line, class, status, citation, last-changed commit — is checked in and regenerated in CI.

8 What this proposal does not resolve

No literature fixes a staleness duration. Any citation offered for the value 3600 is mislabelling a preference as a result.

The calibration question is open, not solved. §3.1 specifies how to *compute* a defensible confidence; it does not say what this system’s prior ought to be. That is a domain judgement about agent-generated claims and belongs to the maintainers.

The power figures are model-based. They assume roughly normal per-observation noise. Whether they transfer to the heavy-tailed metric distributions typical of deep-learning benchmarks is not established, and the worked example in §4.2 inherits that assumption. The appropriate response is simulation from the system’s own observed trial distributions, not a different textbook formula.

One prior sub-test failed and is reported as failed. A measurement of how often a null arm mean

lands on exactly zero returned 0.00% and is uninformative by construction, because continuous draws do not average to exactly zero. The degenerate case arises from bounded or discrete metrics. That sub-test has not been rebuilt.

References

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